

Day 17: Supply Chain Risk Management & PFMEA Framework

Project Author: **Jagriti Singh** | Role: Lead Quality & Process Improvement Engineer

1. Operational Context & Project Objectives

In fast-moving e-commerce setups like **NYKAA**, beauty and personal care supply chains constantly face risks like leaking containers, melted cosmetics, broken glass packaging, and dispatching wrong shades. These issues directly hurt profit margins through scrap costs and customer returns. I designed this case study to map out potential failure points across **NYKAA**'s supply network, restructure inventory flows, and build a practical quality control loop.

2. PFMEA Risk Mapping & RPN Clustering

I evaluated major supply chain breakdown points across **NYKAA**'s fulfillment centers using Severity (S), Occurrence (O), and Detection (D) scales to calculate Risk Priority Numbers (RPN = S x O x D):

Cluster / Operational Failure Mode	Root Cause Drivers	S	O	D	RPN
Cluster 1: Primary Inventory Loss (Heat-Induced Spoilage & Inbound Leakage)	High warehouse temperatures and improper container sealing from suppliers during transit.	9	8	7	504
Cluster 2: Customer Delivery Experience (Wrong SKU Dispatch & Glass Breakage)	Manual picking mix-ups at packing stations and insufficient protective wrap for fragile items.	8	7	6	336
Cluster 3: Last-Mile Delivery Delays (Transit Lags & Order Rejections)	Regional courier bottlenecks leading to missed delivery windows and customer order cancellations.	6	5	4	120

3. Inventory Optimization: ABC Analysis & JIT Integration

To cut down stock holding risks and keep high-demand inventory safe across **NYKAA**'s warehouses, I combined two essential inventory methods:

- ABC Inventory Categorization:** Fast-moving, trending beauty products on **NYKAA** are assigned to Class A for rigorous daily checks. Baseline staples go into Class B, while slow-moving SKUs sit in Class C, ensuring our inspection efforts target high-risk inventory.
- Just-In-Time (JIT) Replenishment:** Stocking high-volume items closer to real-time order signals reduces warehouse storage duration, preventing heat damage to sensitive creams and lipsticks.

4. CAPA Execution & Daily Audit Control Loops

To directly address **NYKAA**'s high-RPN failure points, I structured targeted Action Plans alongside daily operational checks:

- Fixes for Cluster 1:** Added incoming inspection checklists for supplier shipments and set up dedicated climate-controlled rack zones in warehouses.
- Fixes for Cluster 2:** Mandated barcode scanners at packing counters to stop wrong shade dispatches, and switched to shock-absorbent molded pulp trays for fragile glass bottles.
- Daily Dispatch Audits:** Conducting spot checks on daily outgoing batches helps catch wrong SKU dispatches immediately, stopping repeating errors before items leave the facility.

5. Continuous Improvement Strategy: PDCA & Kaizen

- Plan:** Pinpoint critical risk areas in **NYKAA**'s supply chain, assign functional ownership, gather operational error logs, and set clear RPN target reductions.
- Do:** Roll out barcode validation, temperature checks, and ABC-JIT controls in pilot warehouse centers.
- Check:** Review daily dispatch audit findings and tracking logs to measure real reduction in RPN levels.
- Act:** Turn working fixes into standard operating procedure (SOP) documents and keep refining processes through regular Kaizen reviews.

Core Theories, Quality Tools & Methodologies Applied

Tools & Methods: Process FMEA (PFMEA), Risk Priority Number (RPN) Calculation, ABC Classification Analysis, Just-In-Time (JIT) Pull System, Daily Dispatch Audits, Barcode Verification, and CAPA Frameworks.

Underlying Theories & Philosophies: Deming's PDCA Continuous Improvement Cycle (Plan-Do-Check-Act), Kaizen Philosophy (Ongoing Incremental Upgrades), and Lean Waste Elimination (Muda of Scrap, Defect, and Rework).